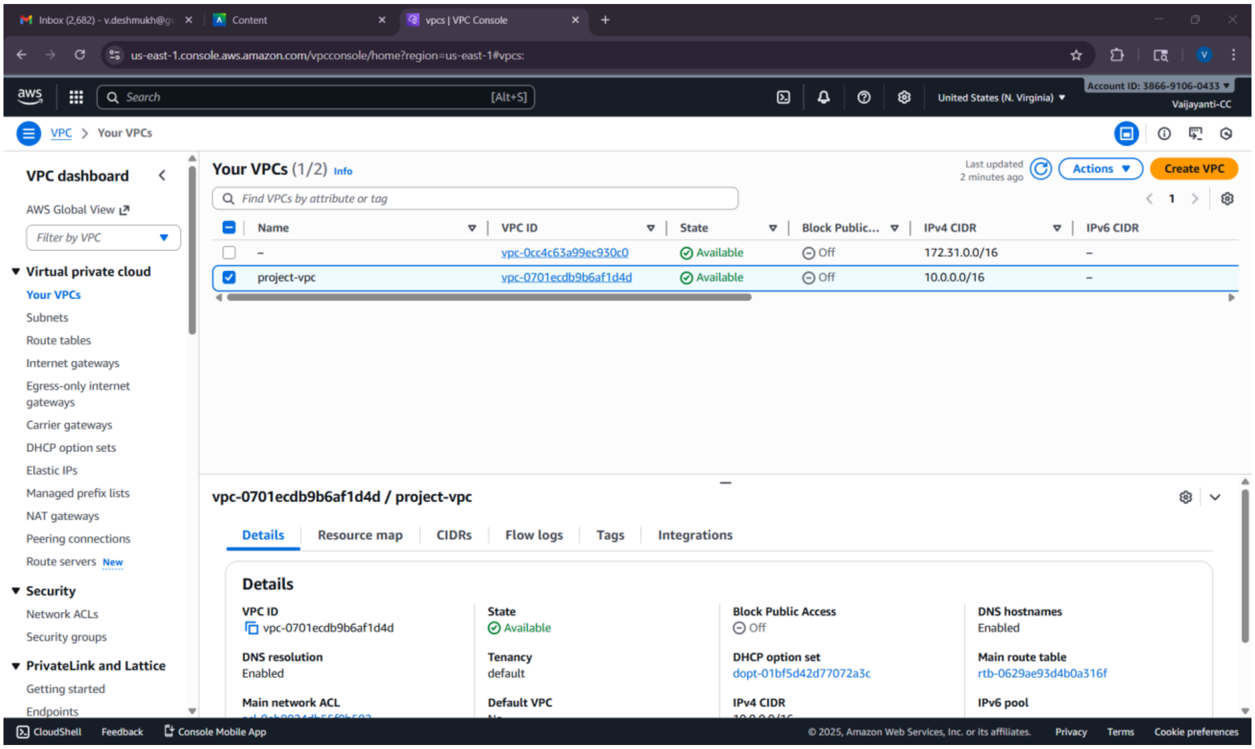
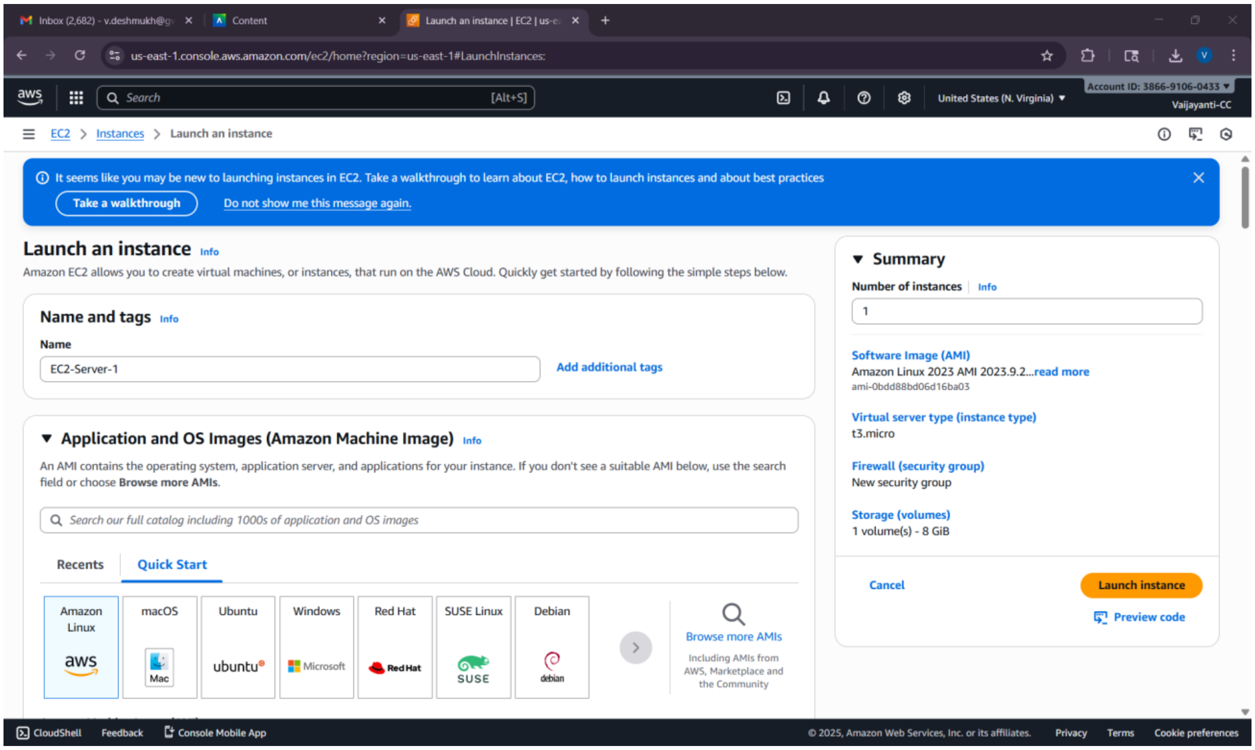
A6 – Elastic Load Balancing

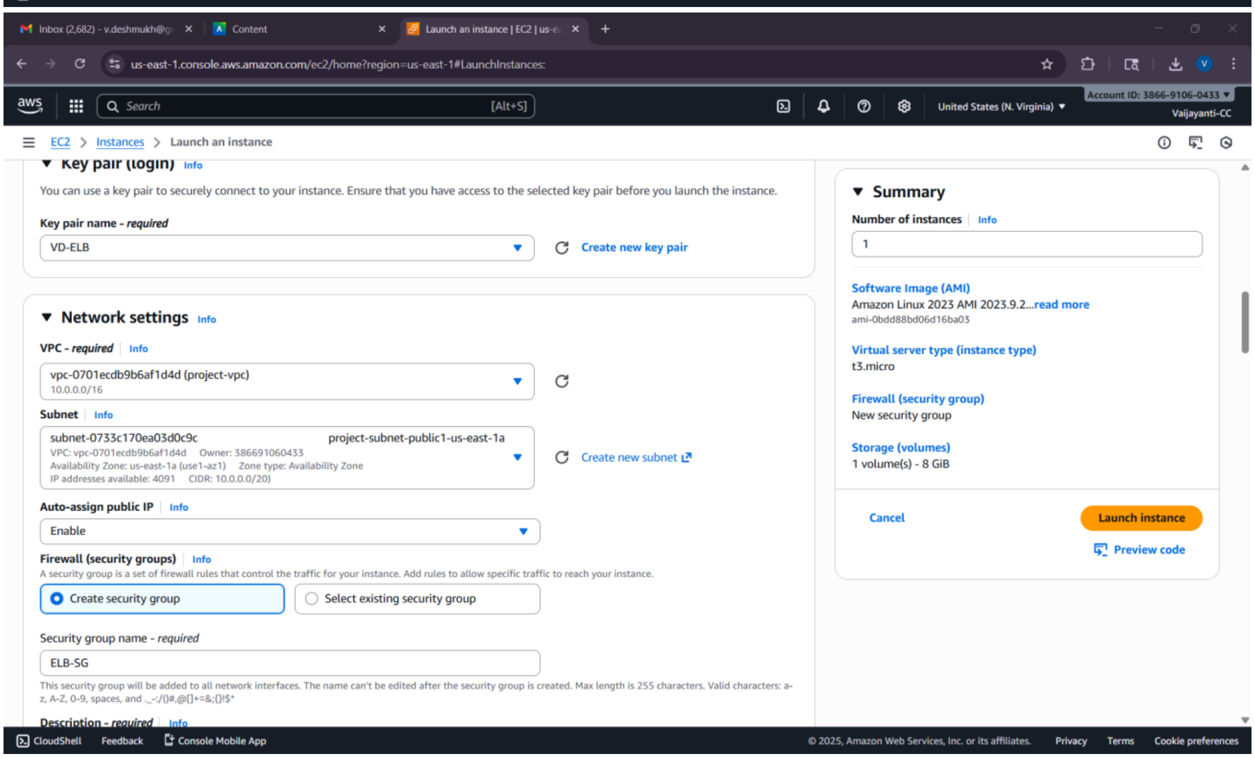
Group 1

1. Reuse the VPC configuration you created in your VPC assignments or use VPC Default

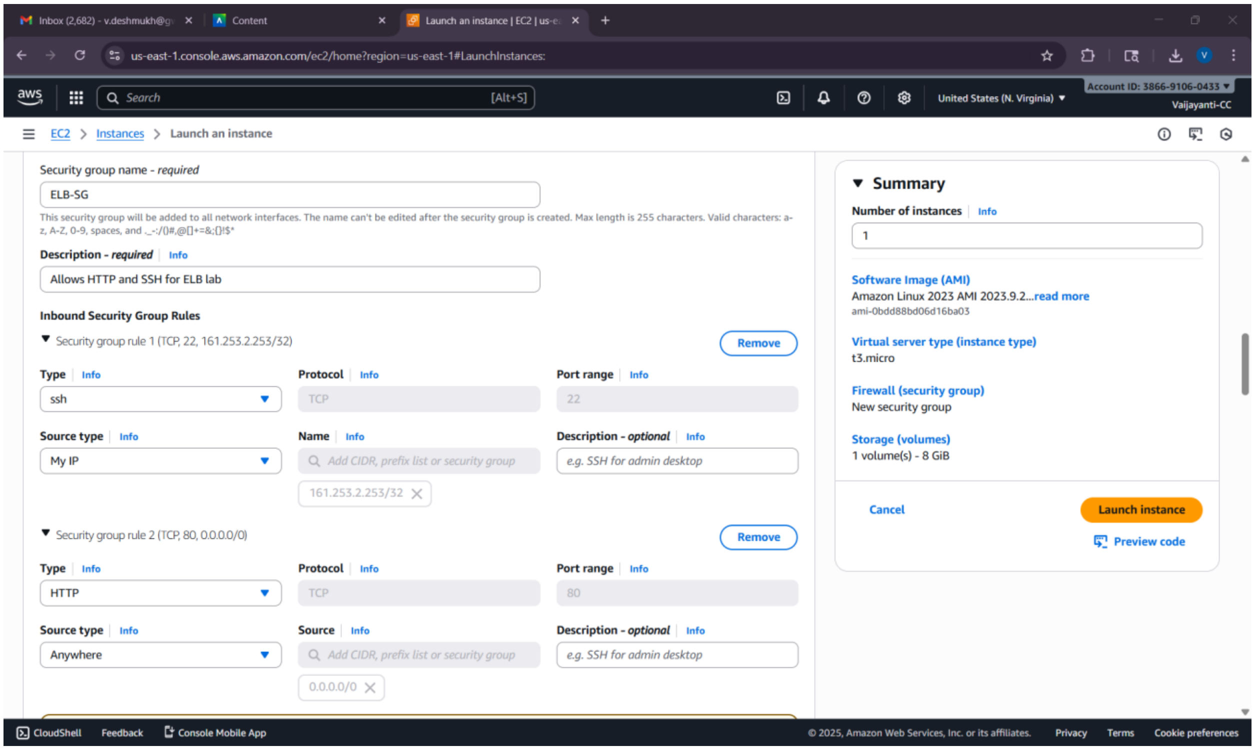


2. Launch 2 EC2s in 2 different AZs and install, start, and enable Apache
   Creating an EC2 instance called as EC2-Server-1

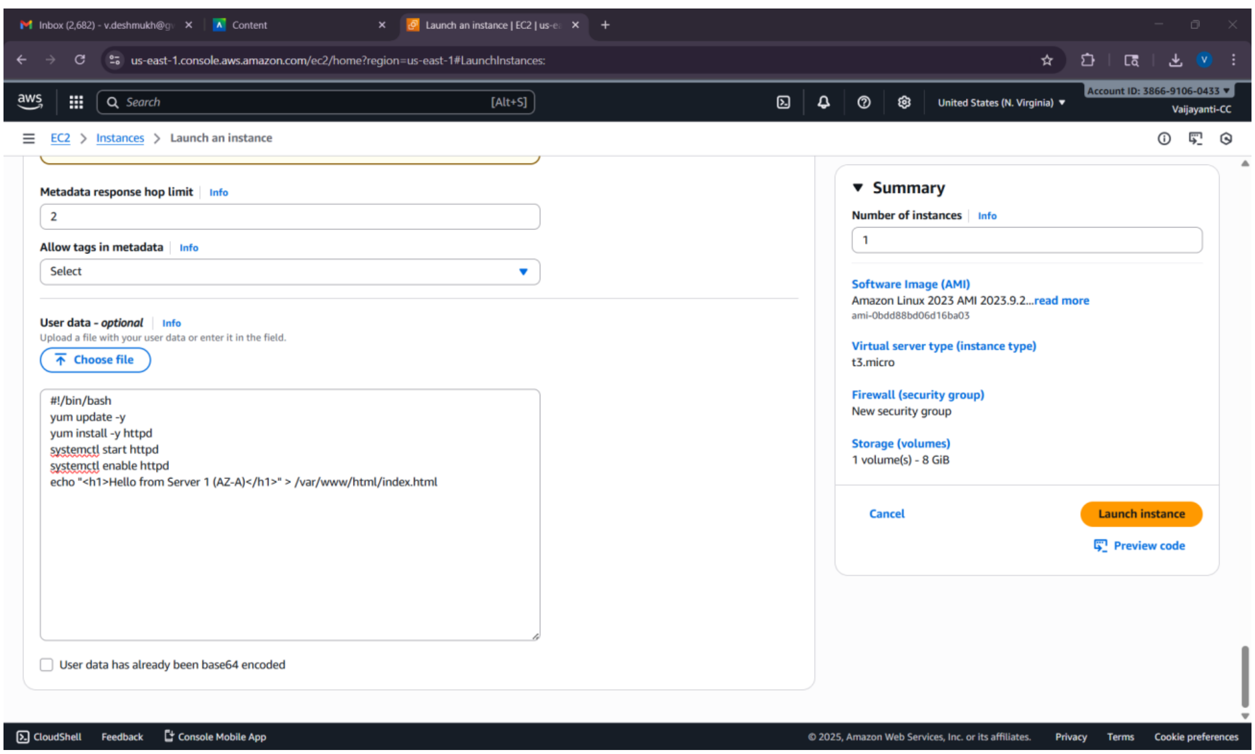




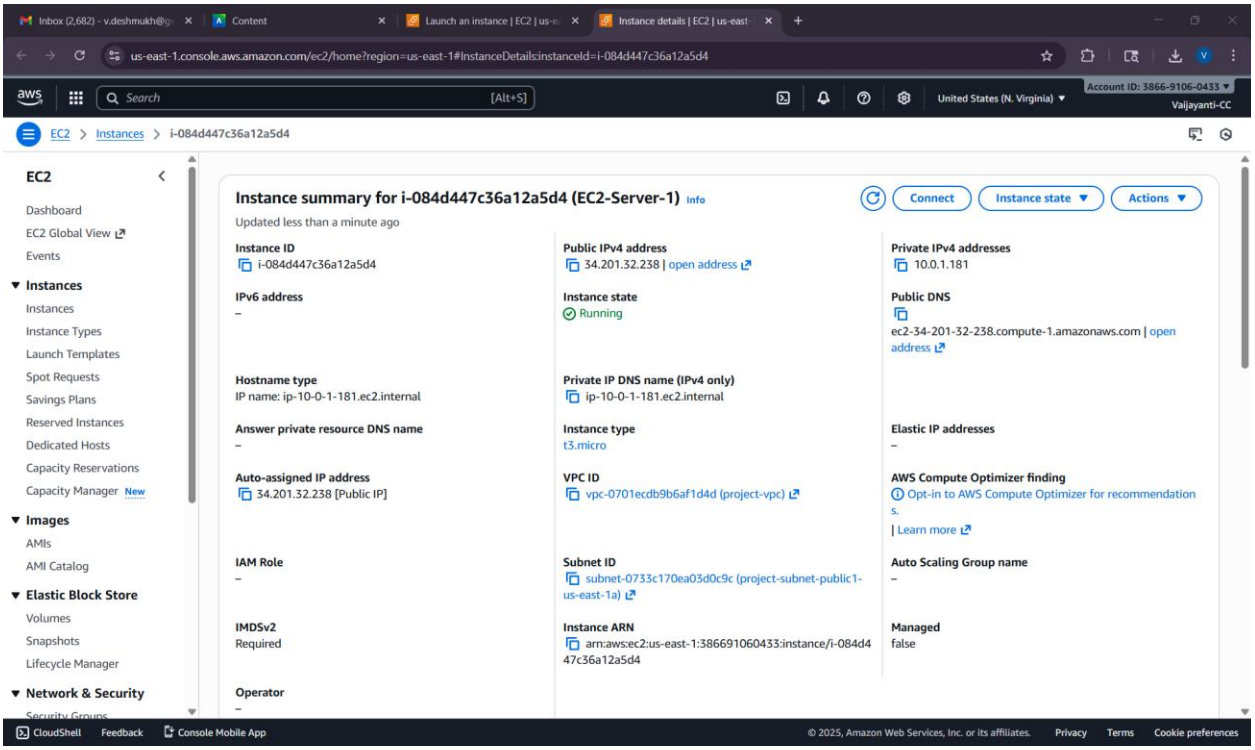
Creating SG with the necessary rules



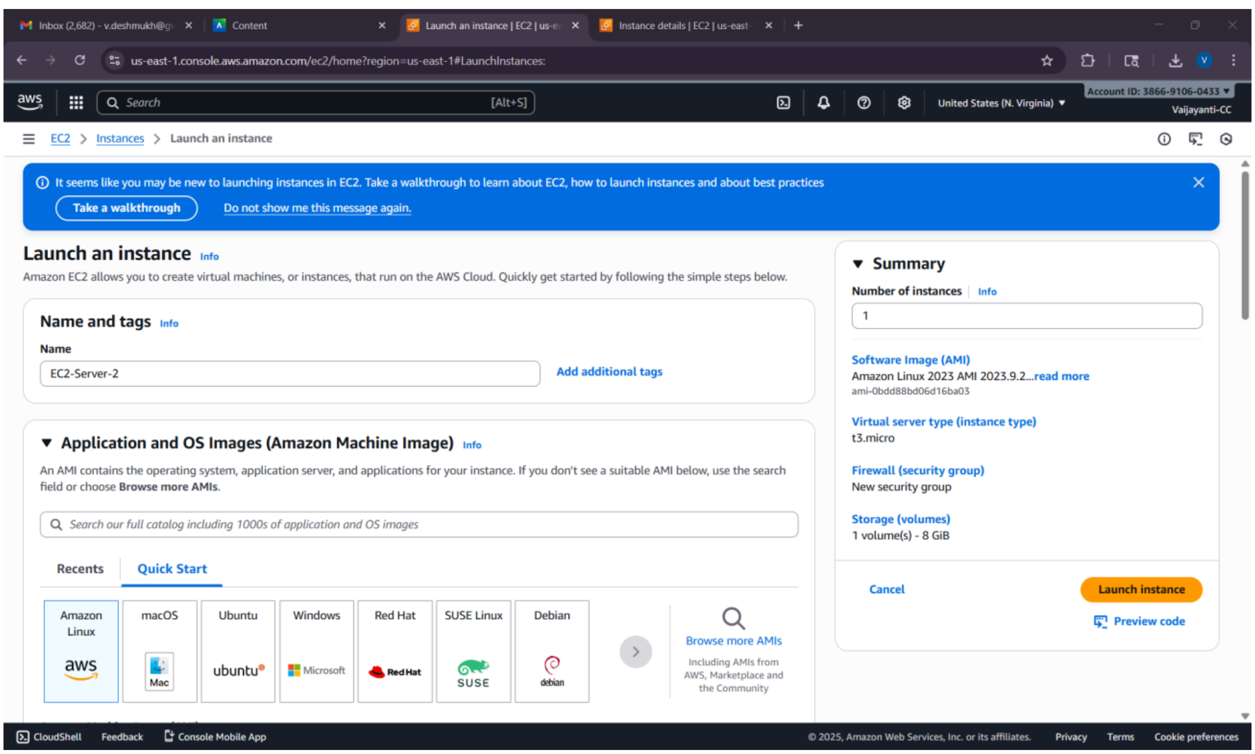
In the advanced details section, in the **"User data"** box, I have pasted the following script so that it will automatically install and start Apache and create a test page.



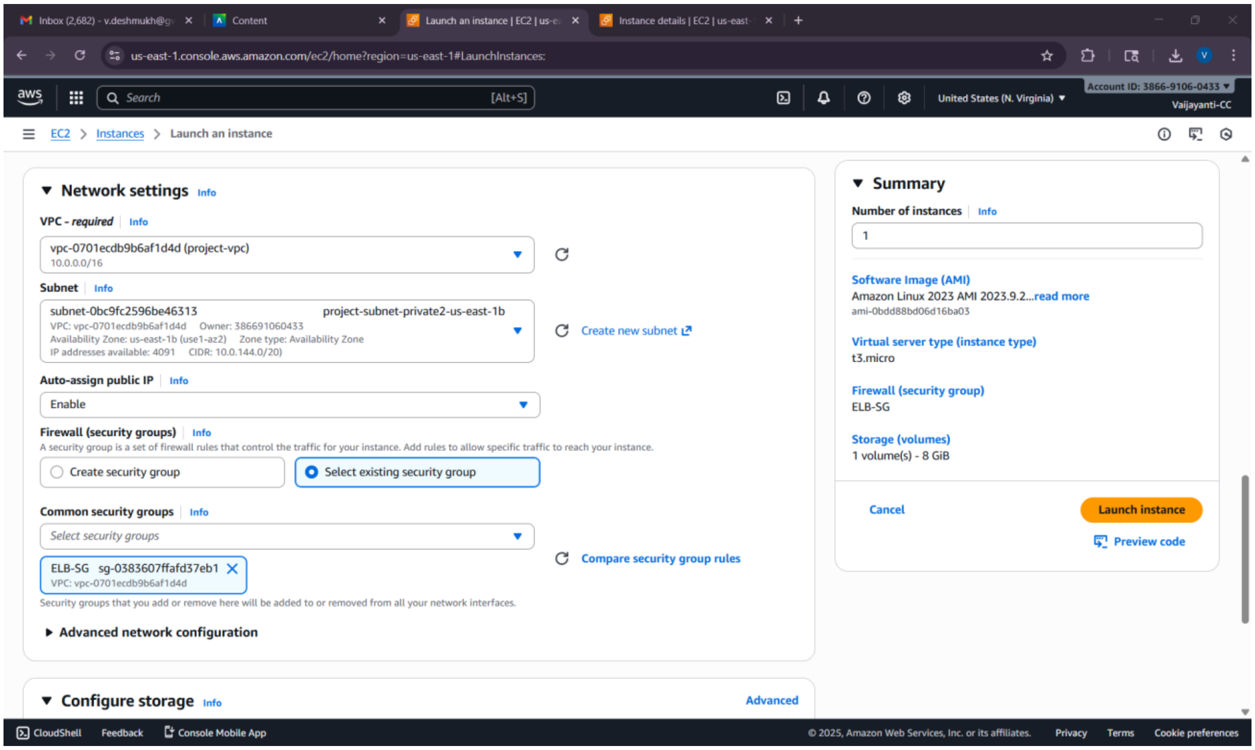
Successfully launched EC2-Server-1 instance



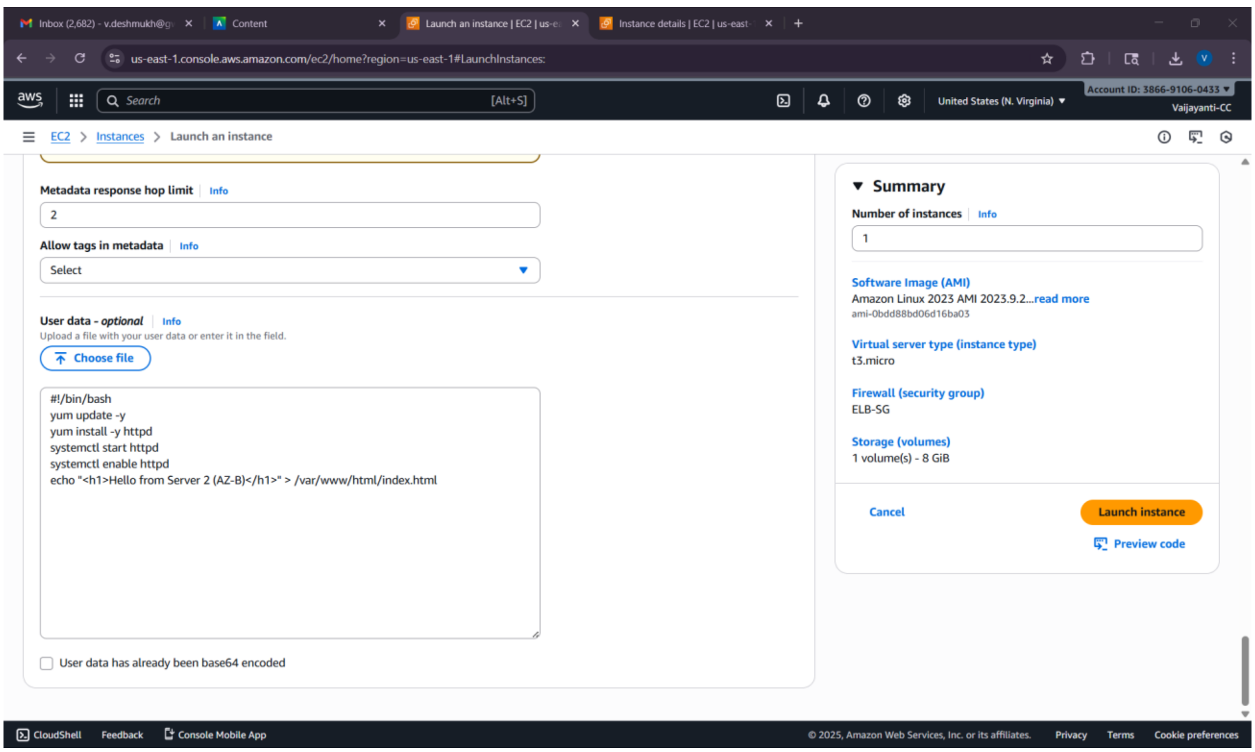
Following the similar steps to create another EC2 instance in a different AZ

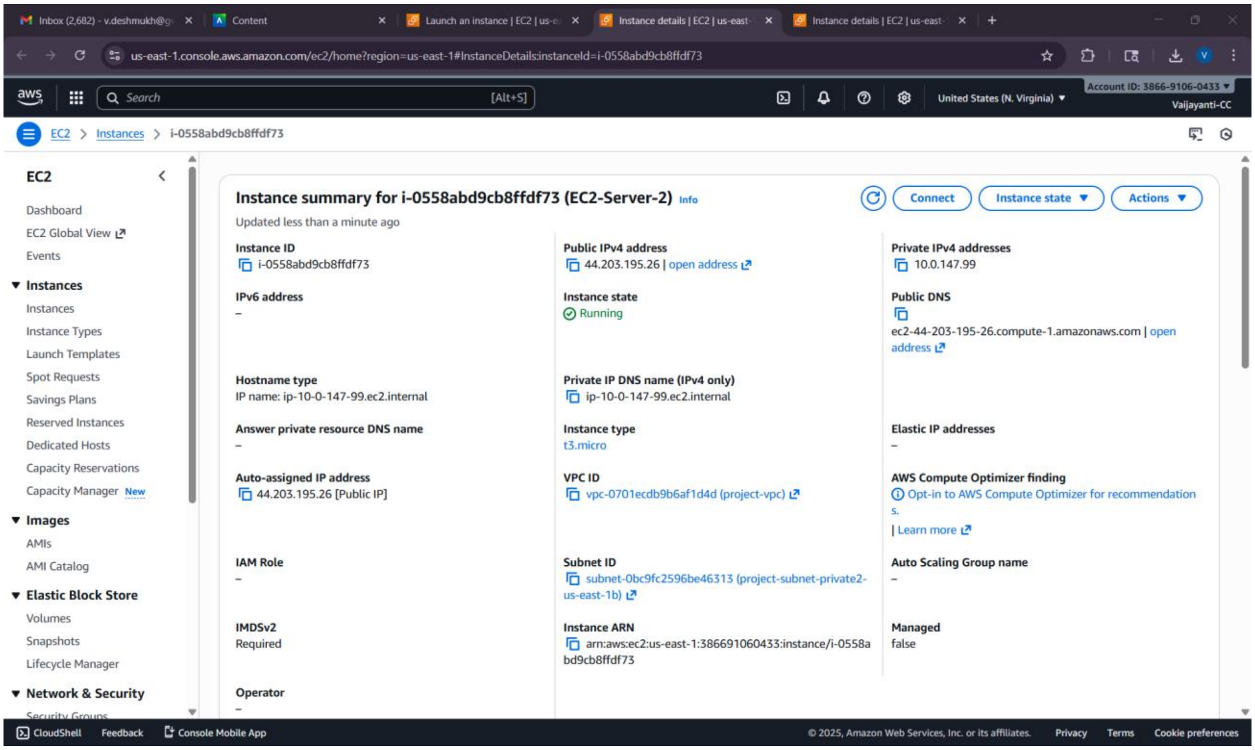


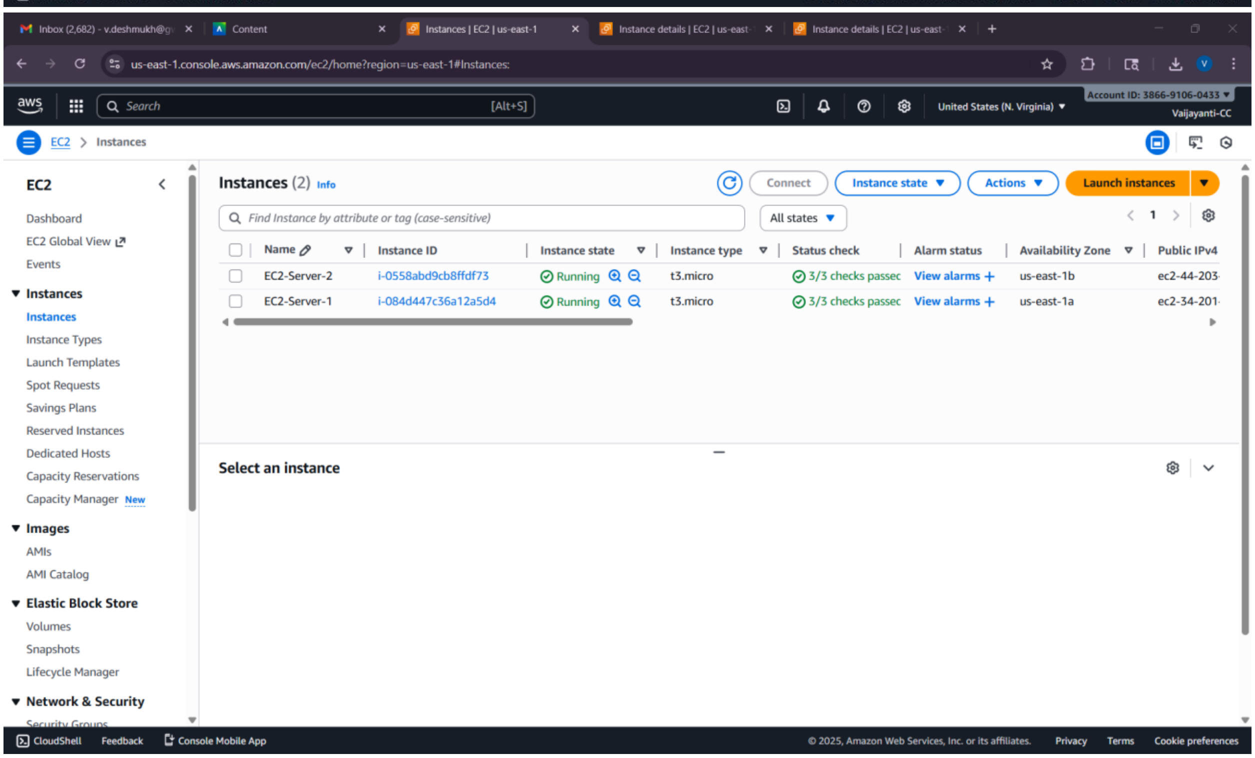
Selecting a subnet that is in different AZ (us-east-1b)



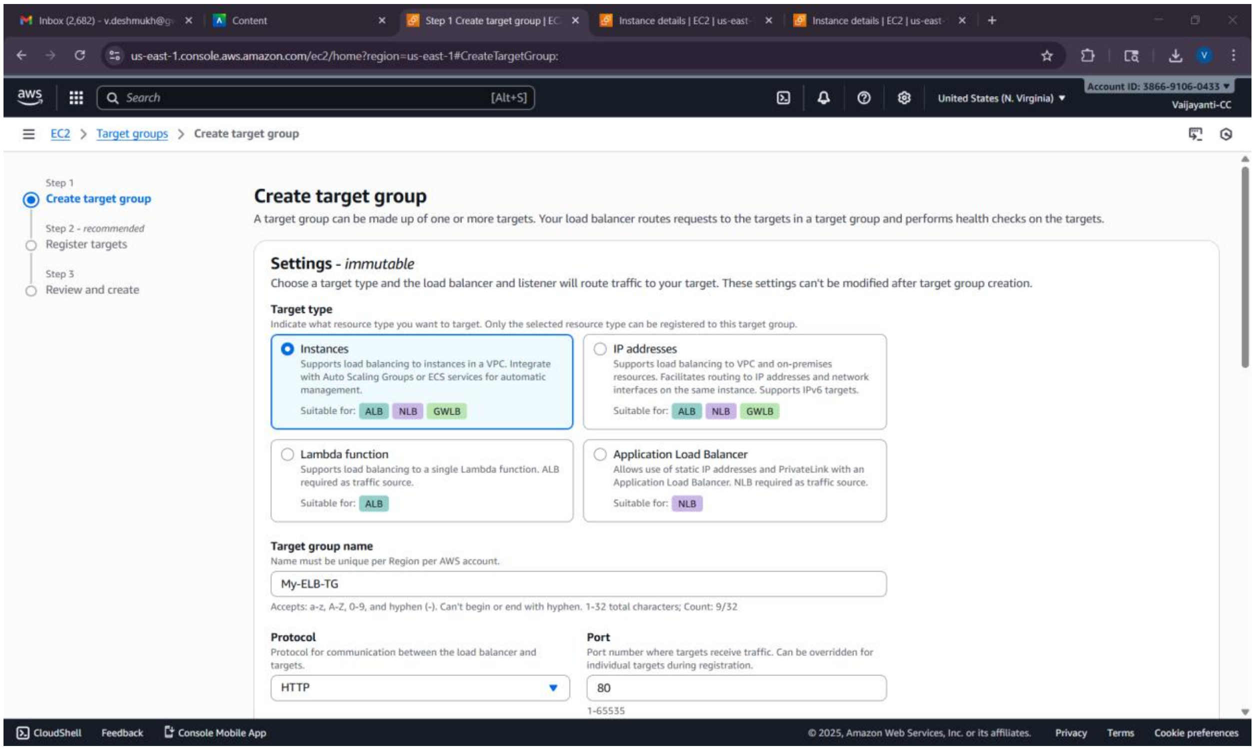
same script, but changing the HTML line so that we know we're hitting the second server

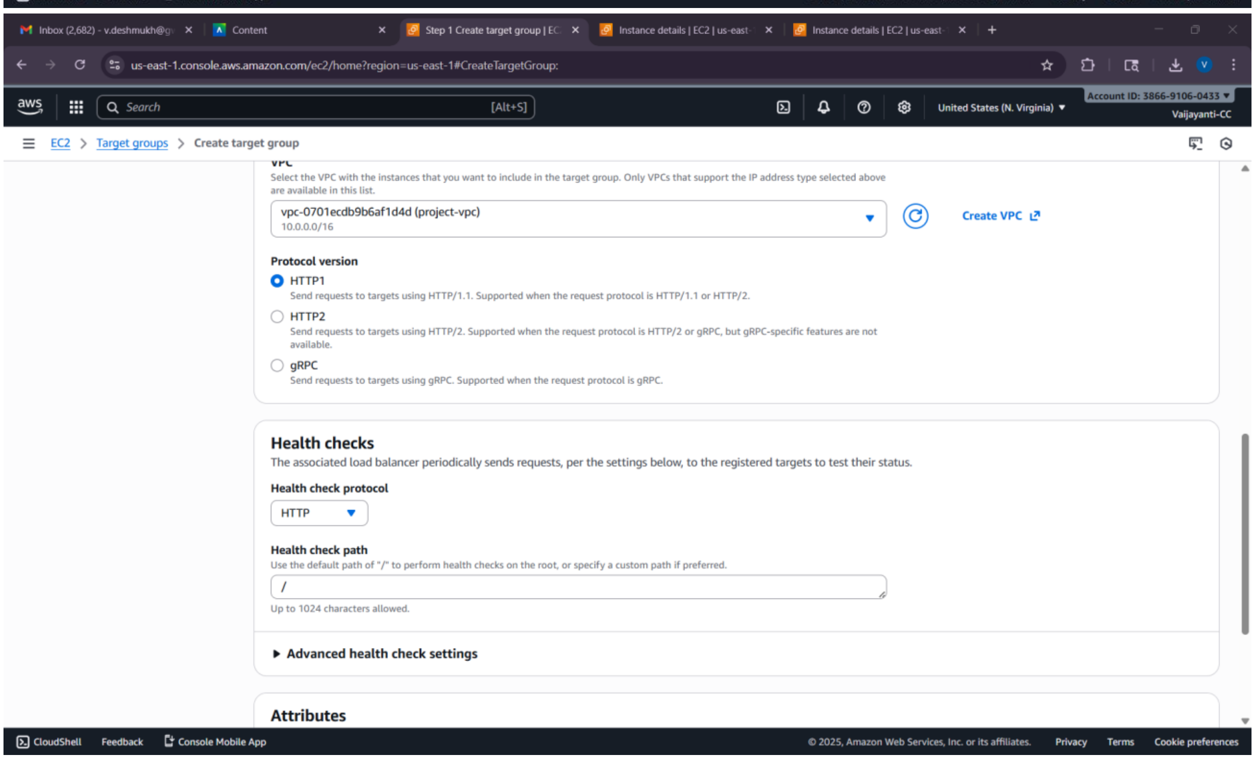


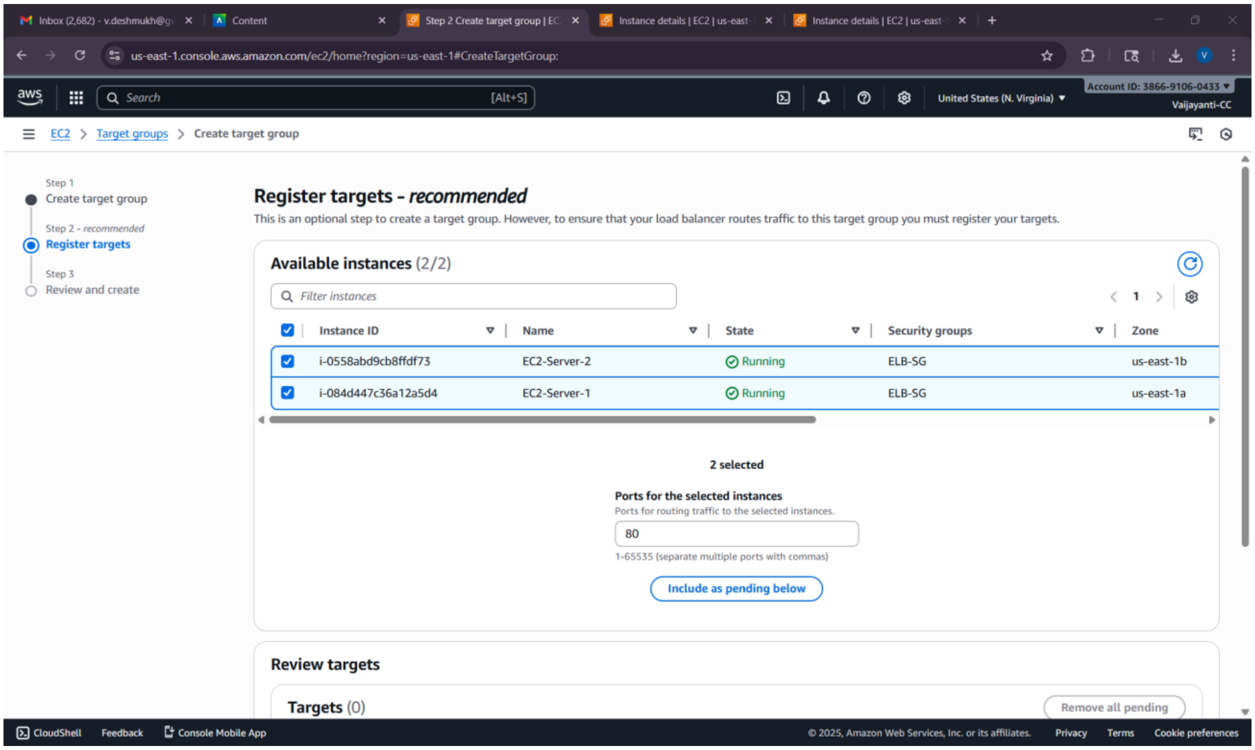


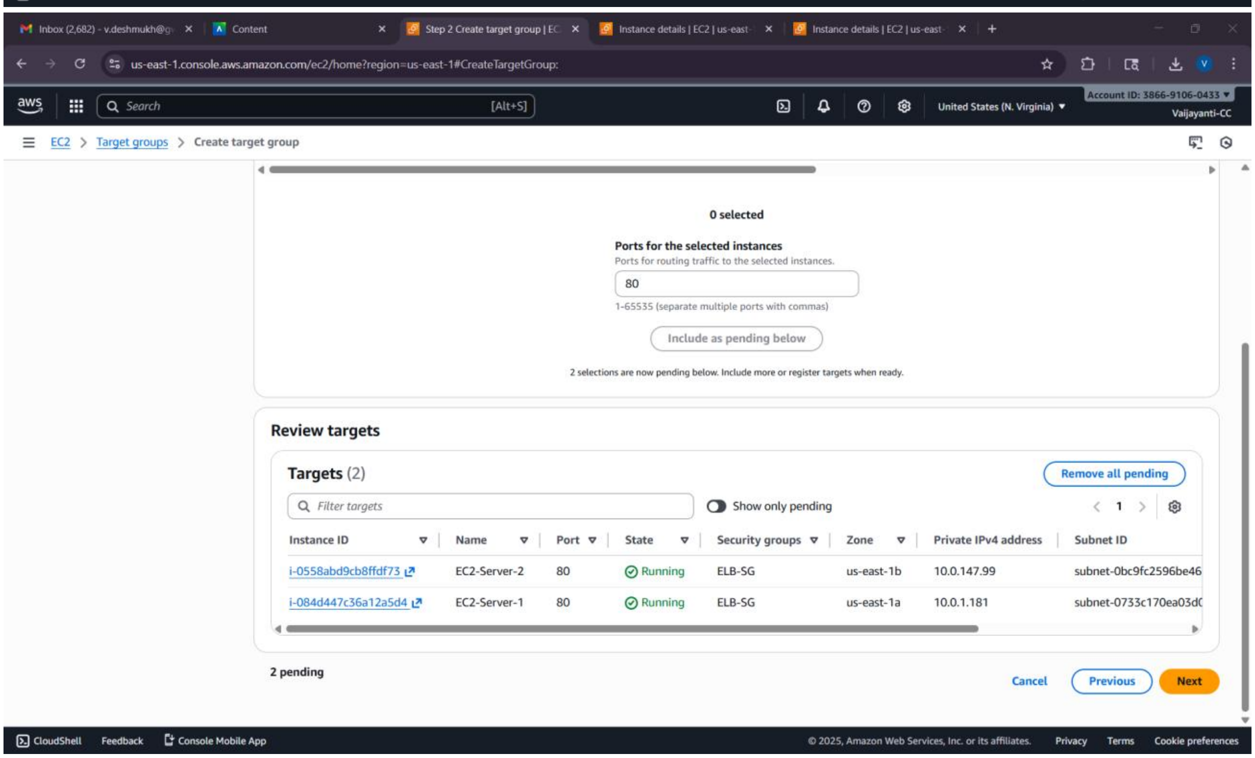


3. Create a Target Group and add these EC2s

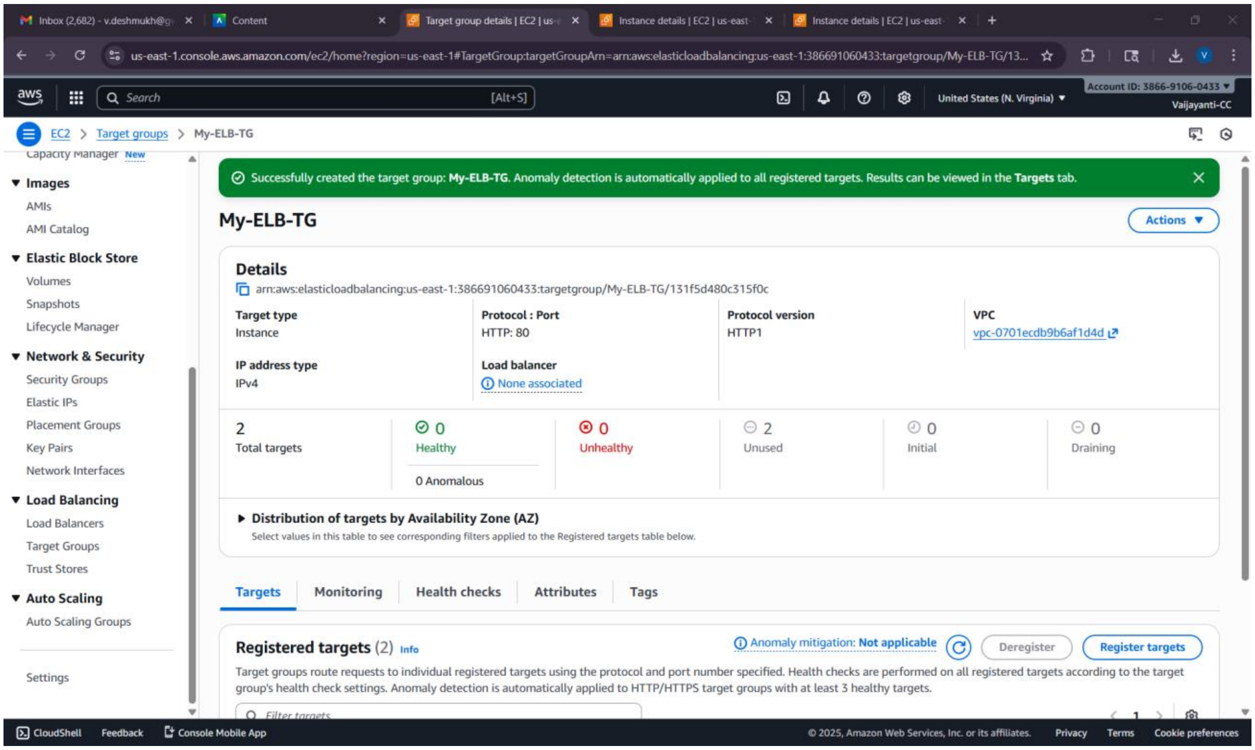

Inbox (2,682) - v.deshmukh@g
Content
Step 1 Create target group | EC
Instance details | EC2 | us-east-
Instance details | EC2 | us-east-
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateTargetGroup:
Search
[Alt+S]
United States (N. Virginia)
Account ID: 3866-9106-0433
Vaijayanti-CC
EC2 > Target groups > Create target group
Step 1
Create target group
Step 2 - recommended
Register targets
Step 3
Review and create
Create target group
A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.
Settings - immutable
Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.
Target type
Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.
Instances
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: ALB NLB GWLB
IP addresses
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: ALB NLB GWLB
Lambda function
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: ALB
Application Load Balancer
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: NLB
Target group name
Name must be unique per Region per AWS account.
My-ELB-TG
Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 9/32
Protocol
Protocol for communication between the load balancer and targets.
HTTP
Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.
80
1-65535
CloudShell
Feedback
Console Mobile App
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

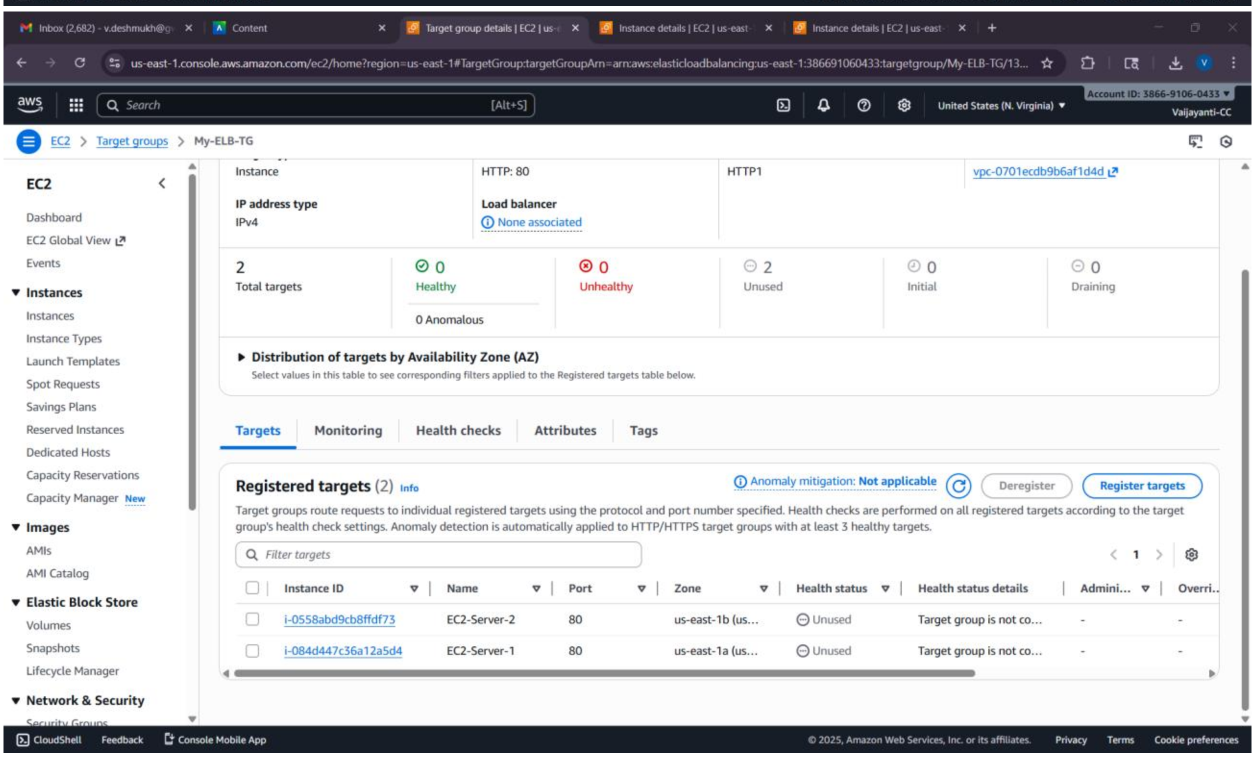

Inbox (2,682) - v.deshmukh@g
Content
Step 1 Create target group | EC
Instance details | EC2 | us-east-
Instance details | EC2 | us-east-
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateTargetGroup:
Search
[Alt+S]
United States (N. Virginia)
Account ID: 3866-9106-0433
Vaijayanti-CC
EC2 > Target groups > Create target group
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.
vpc-0701ecdb9b6af1d4d (project-vpc)
10.0.0.0/16
Create VPC
Protocol version
HTTP1
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.
HTTP2
Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.
gRPC
Send requests to targets using gRPC. Supported when the request protocol is gRPC.
Health checks
The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.
Health check protocol
HTTP
Health check path
Use the default path of "/" to perform health checks on the root, or specify a custom path if preferred.
/
Up to 1024 characters allowed.
Advanced health check settings
Attributes
CloudShell
Feedback
Console Mobile App
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences



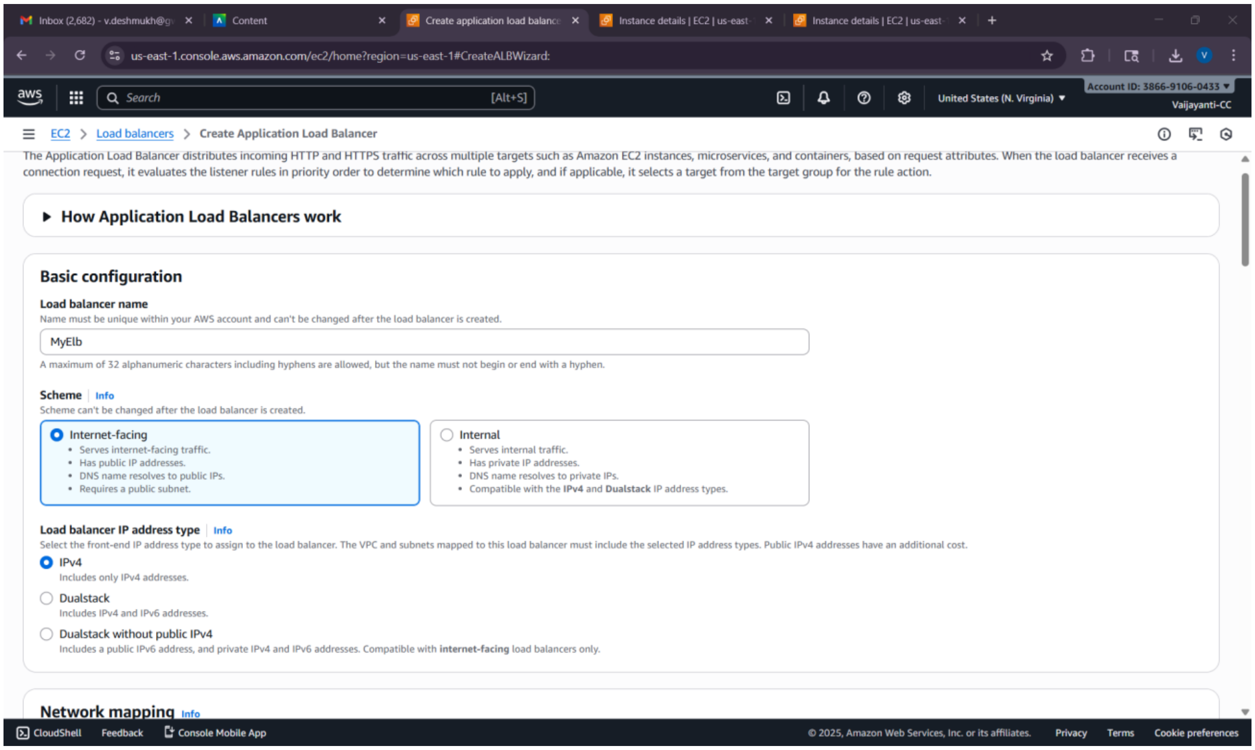


Target Group created





4. Create an ELB, i.e., MyElb, and make sure these 2 EC2s are part of the ELB target group
   Creating an application LB


Inbox (2,682) - v.deshmukh@g
Content
Create application load balance
Instance details | EC2 | us-east-
Instance details | EC2 | us-east-
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateALBWizard:
Search
[Alt+S]
United States (N. Virginia)
Account ID: 3866-9106-0433
Vaijayanti-CC
EC2 > Load balancers > Create Application Load Balancer
The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes. When the load balancer receives a connection request, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.
How Application Load Balancers work
Basic configuration
Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.
MyElb
A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.
Scheme Info
Scheme can't be changed after the load balancer is created.
Internet-facing
Serves internet-facing traffic.
Has public IP addresses.
DNS name resolves to public IPs.
Requires a public subnet.
Internal
Serves internal traffic.
Has private IP addresses.
DNS name resolves to private IPs.
Compatible with the IPv4 and Dualstack IP address types.
Load balancer IP address type Info
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.
IPv4
Includes only IPv4 addresses.
Dualstack
Includes IPv4 and IPv6 addresses.
Dualstack without public IPv4
Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with internet-facing load balancers only.
Network mapping Info
CloudShell
Feedback
Console Mobile App
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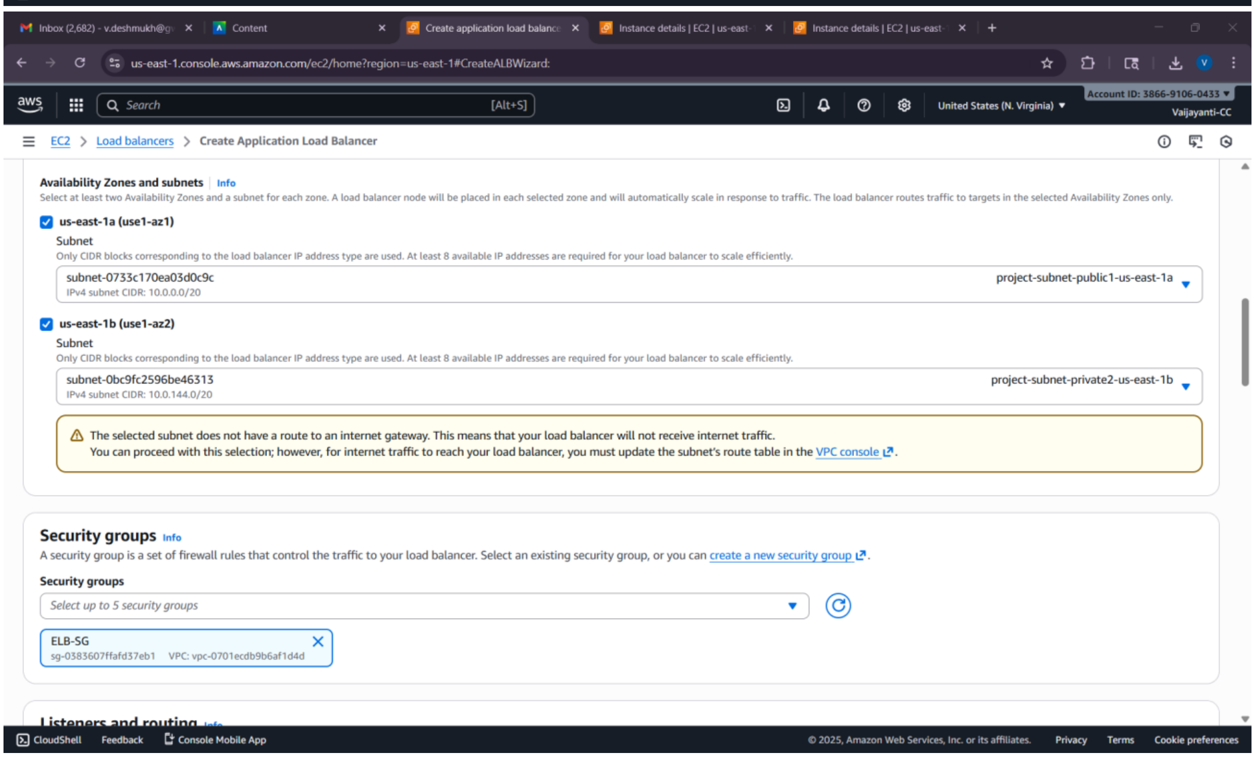
Privacy
Terms
Cookie preferences


Inbox (2,682) - v.deshmukh@g
Content
Create application load balance
Instance details | EC2 | us-east-
Instance details | EC2 | us-east-
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateALBWizard:
Search
[Alt+S]
United States (N. Virginia)
Account ID: 3866-9106-0433
Vaijayanti-CC
EC2 > Load balancers > Create Application Load Balancer
Availability Zones and subnets Info
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.
us-east-1a (use1-az1)
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.
subnet-0733c170ea03d0c9c
IPv4 subnet CIDR: 10.0.0.0/20
project-subnet-public1-us-east-1a
us-east-1b (use1-az2)
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.
subnet-0bc9fc2596be46313
IPv4 subnet CIDR: 10.0.144.0/20
project-subnet-private2-us-east-1b
The selected subnet does not have a route to an internet gateway. This means that your load balancer will not receive internet traffic.
You can proceed with this selection; however, for internet traffic to reach your load balancer, you must update the subnet's route table in the VPC console.
Security groups Info
A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can create a new security group.
Security groups
Select up to 5 security groups
ELB-SG
sg-0383607ffafd37eb1 VPC: vpc-0701ecdb9b6af1d4d
Listeners and routing Info
CloudShell
Feedback
Console Mobile App
© 2025, Amazon Web Services, Inc. or its affiliates.
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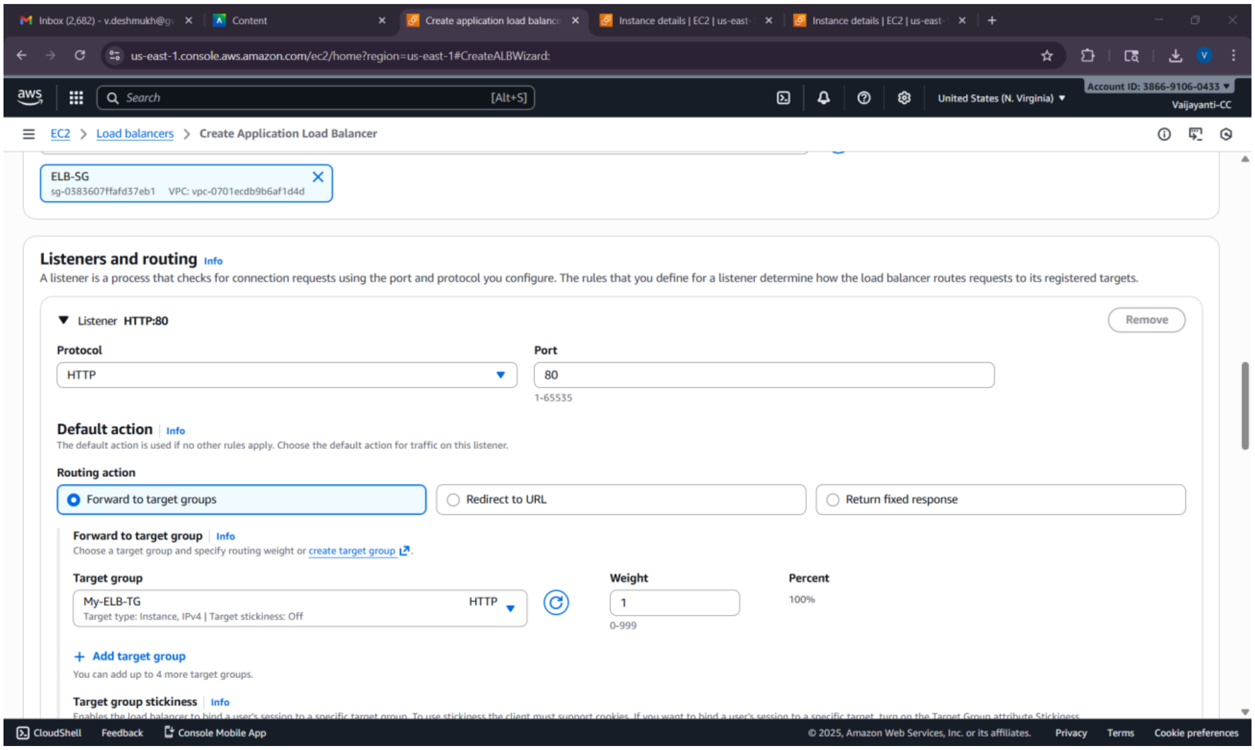
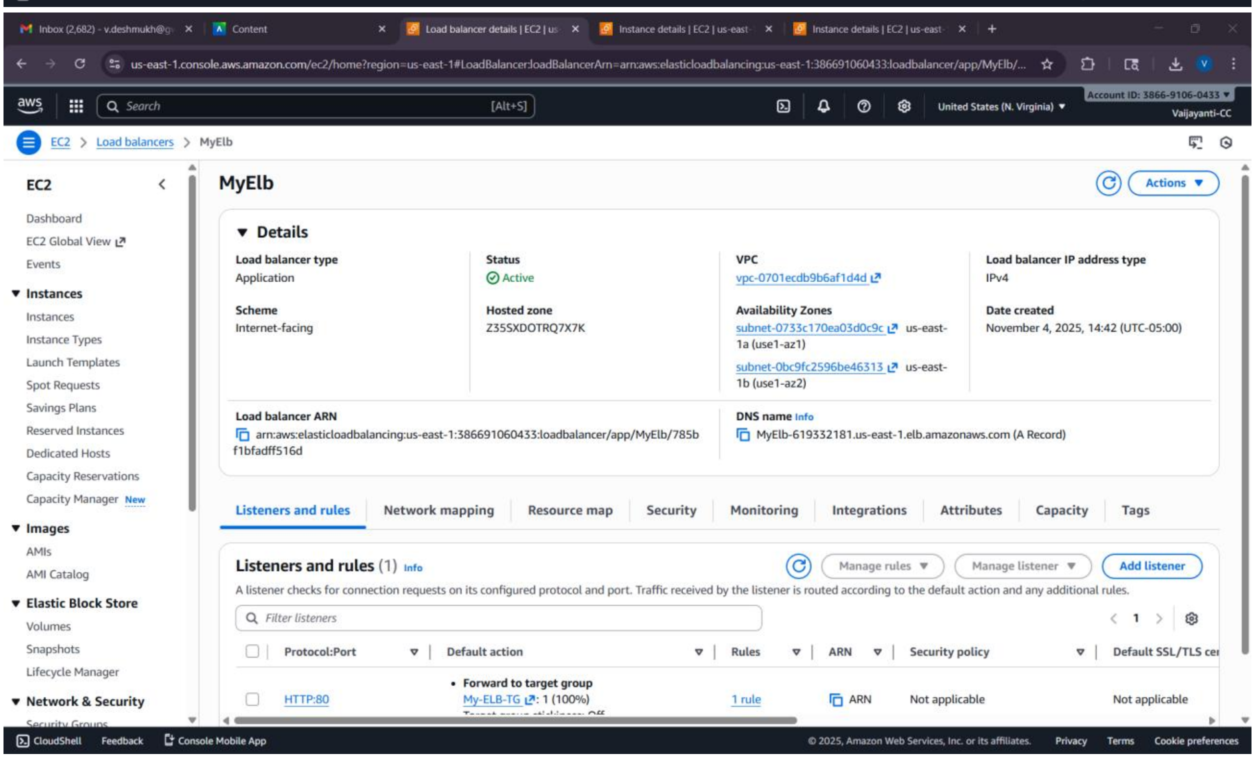
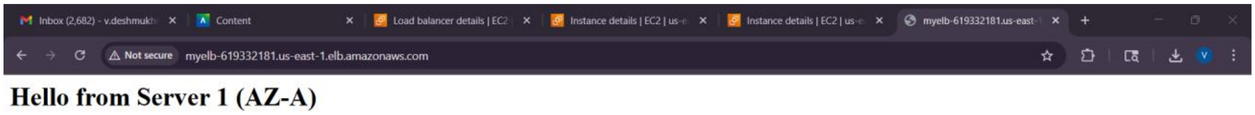
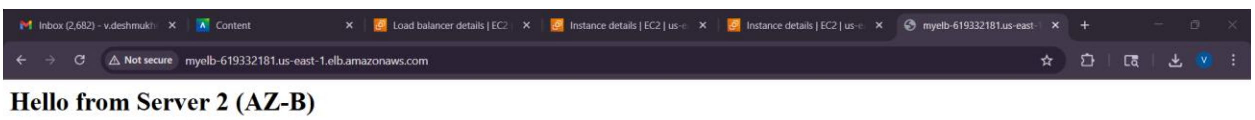




5. Access the ELB via the web
   DNS Name : MyElb-619332181.us-east-1.elb.amazonaws.com

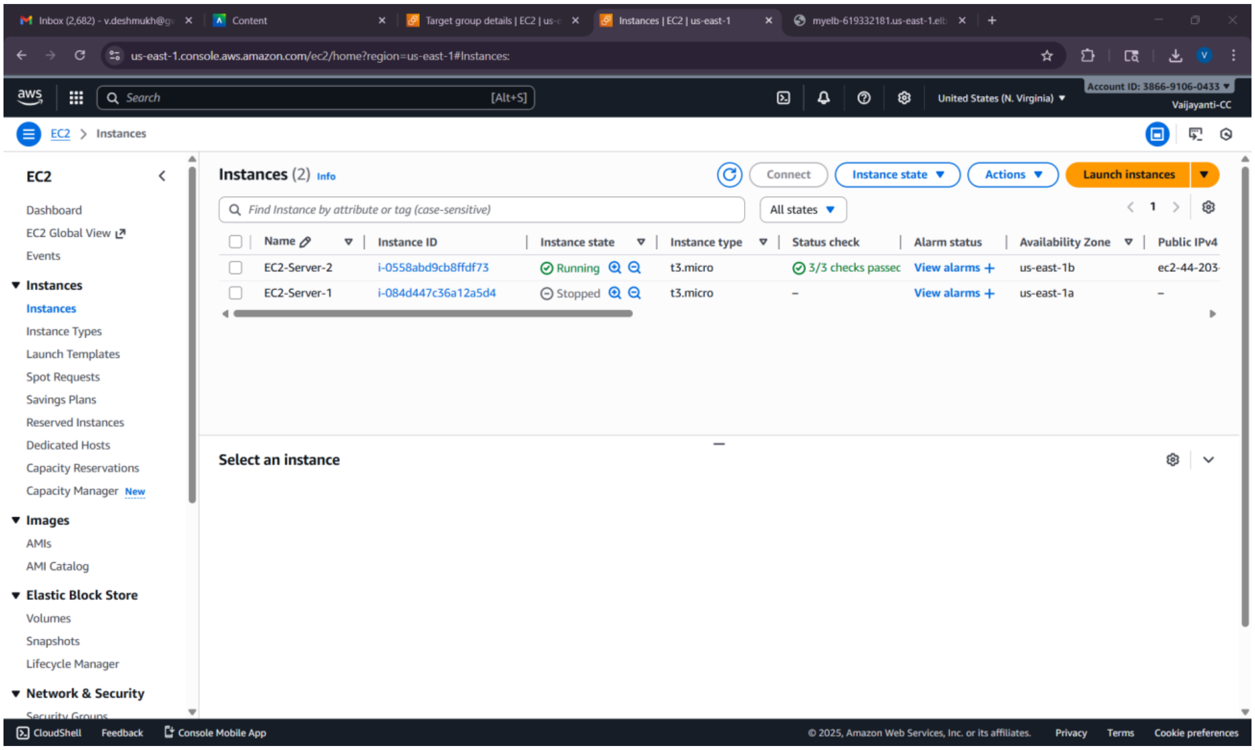


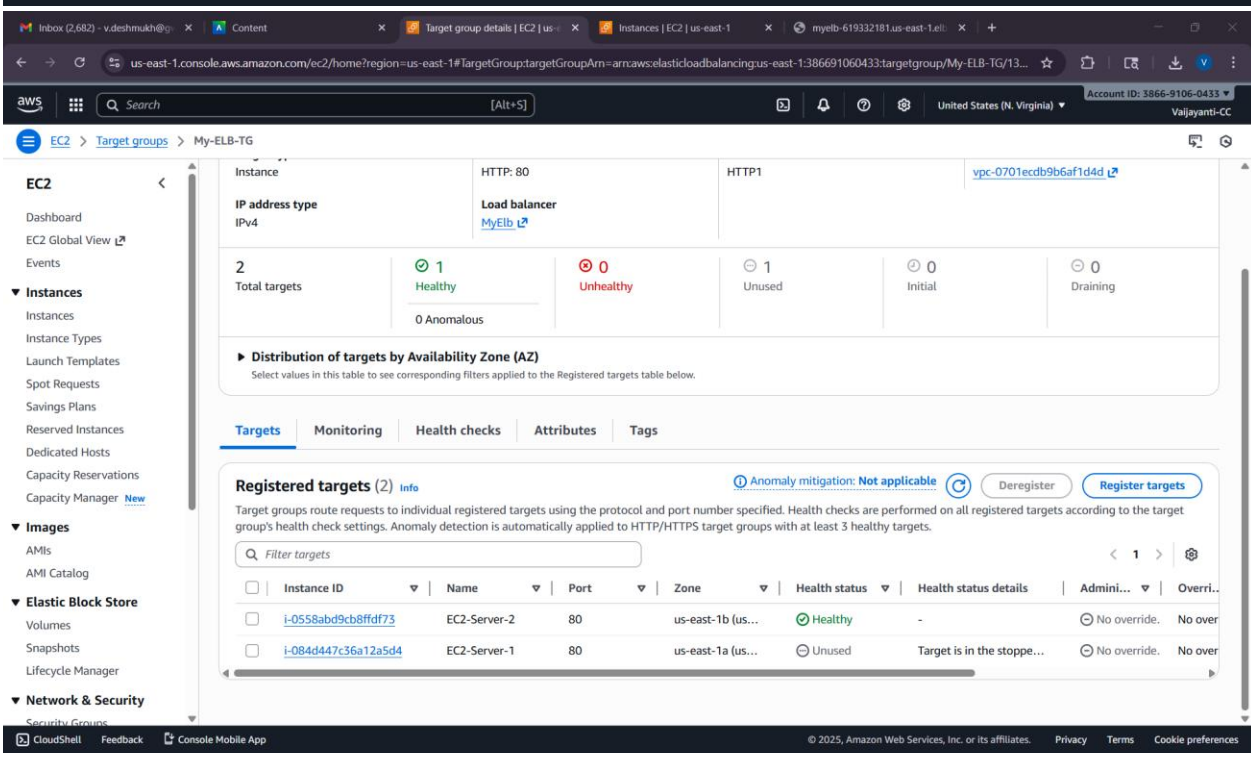
Upon refreshing the page



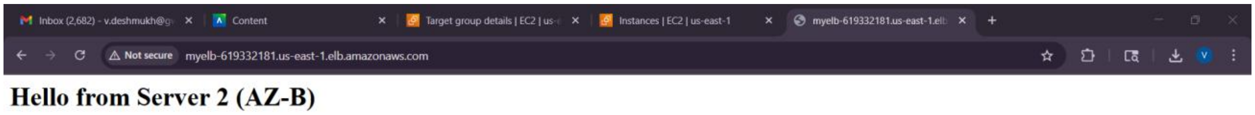
6. Stop an EC2 and access again the ELB via the web

   Stopping EC2-Server-1 instance

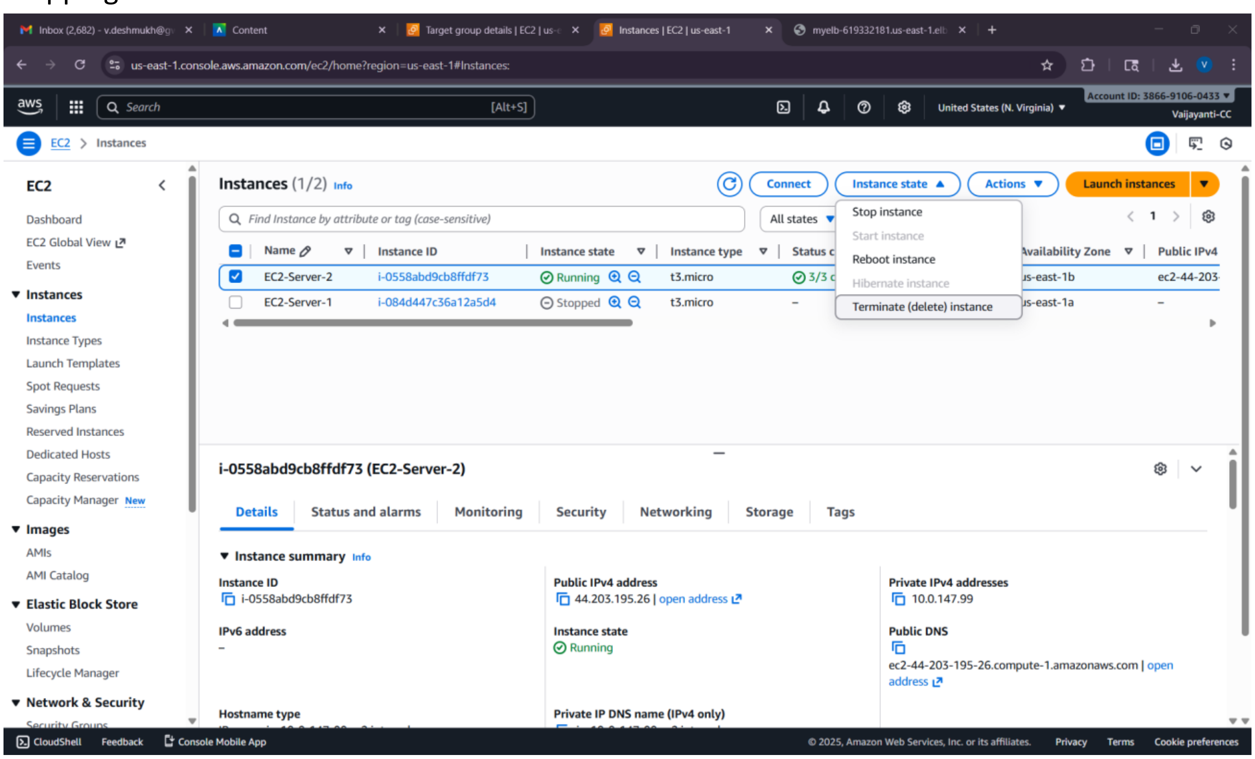


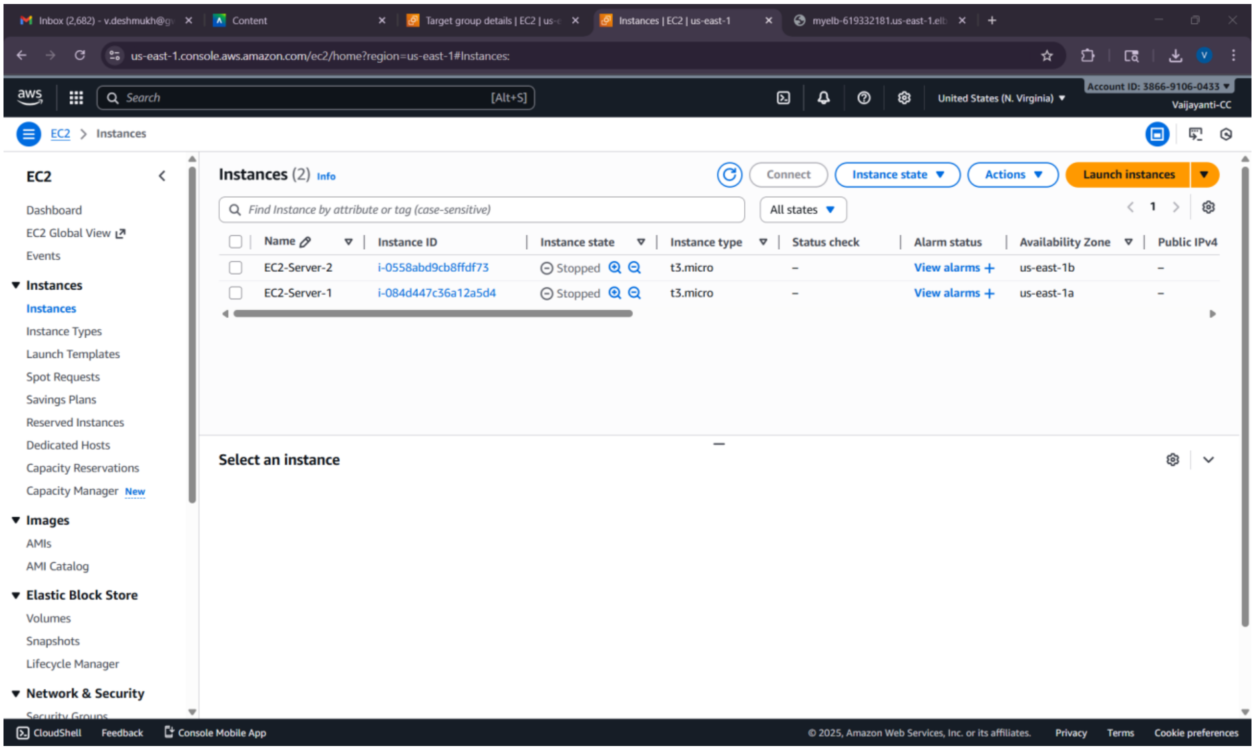


Only “Hello from Server 2 (AZ-B)” can be seen over and over. The load balancer knows Server 1 is stopped, so it automatically stops sending any traffic to it. The website stays online and is served by the other healthy instance.

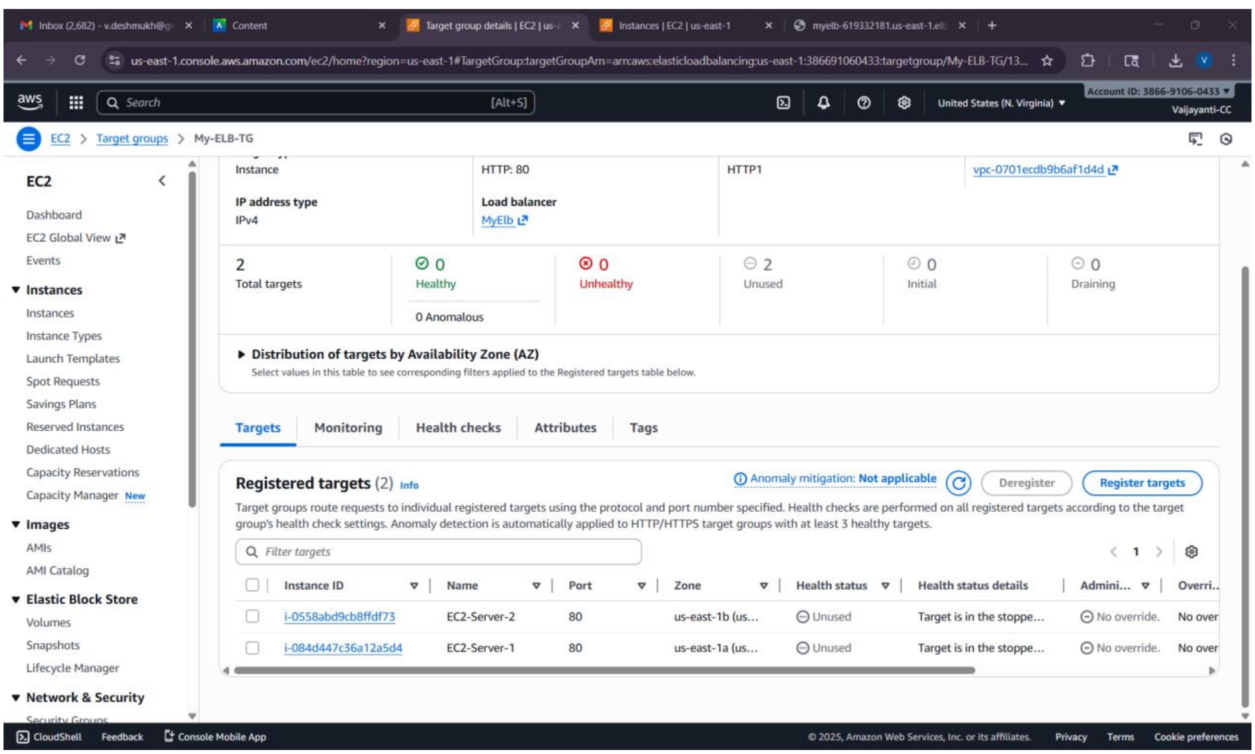


7. Stop the 2nd EC2 and access again the ELB via the web
   Stopping EC2-Server-2 instance

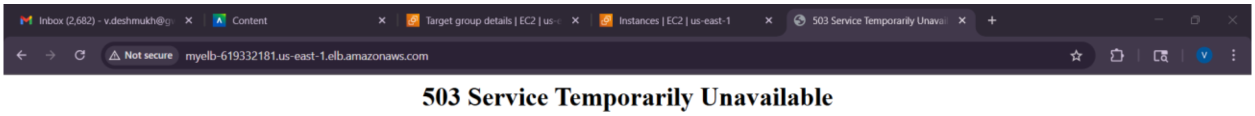




The status of EC2-Server-2 instance in TG changed from ‘healthy’ to ‘unused’



Upon refreshing the website



8. Describe results

The application load balancer successfully distributes the traffic across both ec2 instances. Upon stopping EC2-Server-1, the ELB health checks correctly identified the instance and all subsequent traffic was automatically rerouted to the remaining EC2-Server-2, which consistently displayed the Server 2 (AZ-B) page. This demonstrated the ELB's high availability by seamlessly handling a single-instance failure.
After stopping the second instance, EC2-Server-2, the ELB detected that no healthy targets remained in its target group. Hence when I tried accessing the dns, it failed (503 Service Unavailable)